

DAIRU LIU

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EDUCATION

Nankai University

B.Eng. in Software Engineering; GPA: 3.75

Tianjin, China

Sep. 2023 – Jun. 2027 (expected)

The Affiliated High School of Peking University

Beijing, China

RESEARCH INTERESTS

- **Embodied Intelligence, Spatial Intelligence, Multimodal Reasoning**

PUBLICATIONS

1. **Thinking with Visual Abstract: Enhancing Multimodal Reasoning via Visual Abstraction** [arxiv:2505.20164] [GitHub]
*Dairu Liu**, Ziyue Wang*, Minyuan Ruan*, Fuwen Luo, Chi Chen, Peng Li, Yang Liu
ICML Under Review, NeurIPS 2025 Multimodal Algorithmic Reasoning Workshop

RESEARCH EXPERIENCE

Research Intern, GALBOT | IIIS, Tsinghua University

supervised by Prof. Li Yi; collaborated closely with Zekun Qi

Beijing, China

Dec. 2025 – now

- **General Reward Model for Humanoid Robot**

Research Intern, Institute for AI Industry Research (AIR), Tsinghua University

supervised by Prof. Peng Li, Prof. Yang Liu; collaborated closely with Ziyue Wang

Beijing, China

Apr. 2024 – Dec. 2025

- **Multimodal Reasoning:** Proposed **Visual Abstract Thinking (VAT)**, a novel reasoning paradigm inspired by human visual cognition. VAT enables MLLMs to reason more effectively by leveraging visual abstractions rather than explicit verbal Chain-of-Thought. By guiding models to focus on semantic and geometric concepts, VAT achieves a **17% improvement over the GPT-4o baseline** and consistently outperforms Chain-of-Thought prompting on **visual perception and reasoning tasks**, while requiring **fewer tokens** and delivering **higher performance**.
- **4D Escape-Room Environment for Multimodal Reasoning:** Proposed a customizable 4D (spatio-temporal) escape-room environment to evaluate multimodal/omni LLM agents through free-form exploration, emphasizing active perception and spatio-temporal cross-modal alignment.
- **Self-Evolving Agents:** Exploring how to strengthen agent performance by enabling generalization from weak to strong supervision.

Research Intern, Pervasive Human-Computer Interaction Lab (π Lab), Tsinghua University

supervised by Prof. Chun Yu

Beijing, China

Sep. 2024 – May 2025

- **Psychological Agent:** Implemented a **Cognitive Behavioral Therapy (CBT) expert agent** in a large-model-based AI psychological counseling system deployed on the **CYL 12355 Youth Service Hotline (Fujian)**. The agent adheres to structured CBT protocols, including automatic thought identification, cognitive distortion analysis, and behavior-oriented interventions. [Related Report]

INTERNSHIP EXPERIENCE

Industry Research Intern, Jingxing Capital

Beijing, China

Jan. 2024 – Feb. 2024

- Studied equity research reports and market rotation patterns between large- and small-cap stocks.
- Developed Python scripts that auto-generated the fund's daily NAV report, saving analysts manual effort.
- Observed investment decision workflows and gained first-hand exposure to the finance industry.

AWARDS AND COMPETITIONS

National Scholarship (Undergraduate, 10000CNY)	Sep. 2024
Second Prize , NOIP 2018 (National Olympiad in Informatics in Provinces)	Dec. 2018
Second Prize , CSP-S 2019 (CCF Certified Software Professional)	Dec. 2019

EXTRACURRICULARS

Vice President, Quantitative Finance & Investment Society, Nankai University	Sep. 2025 – Present
Member, University Ski Team, Nankai University	2024 – 2025
Academic Department Vice-Head, Psychology Association, Nankai University	Sep. 2024 – Sep. 2025

SKILLS

Programming: C++, Python, MATLAB, $\text{\LaTeX}$

Languages: Mandarin Chinese (native); English (IELTS 7.0 – L 7.5, R 8.0, S 6.0, W 6.0; CET-6 608)

Interests: Skiing, Psychology, Trading